

OOP with Intelligent Systems

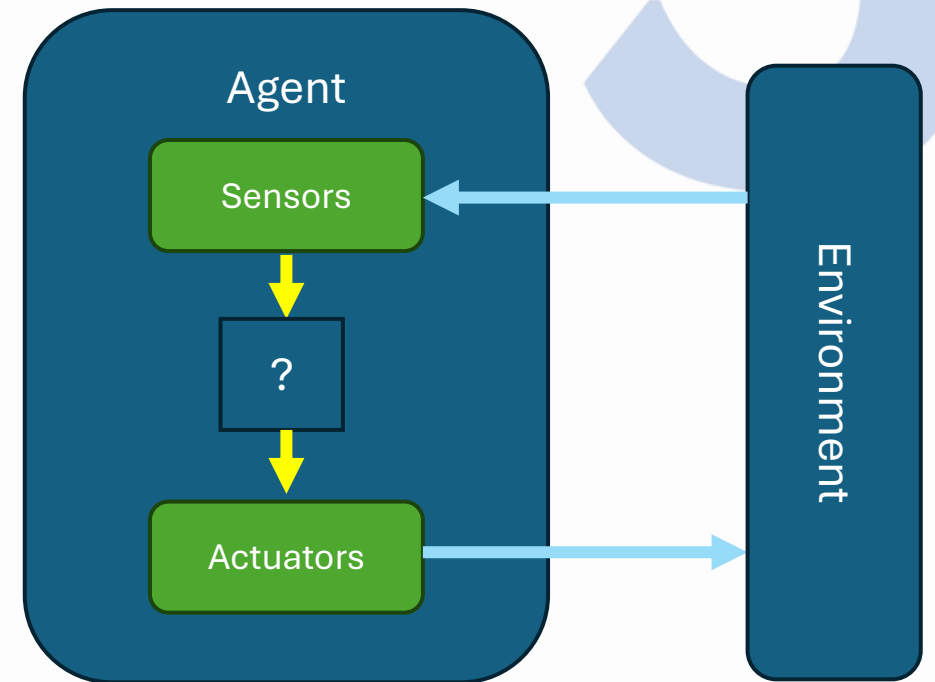
DS205.3 – Data Science in Python

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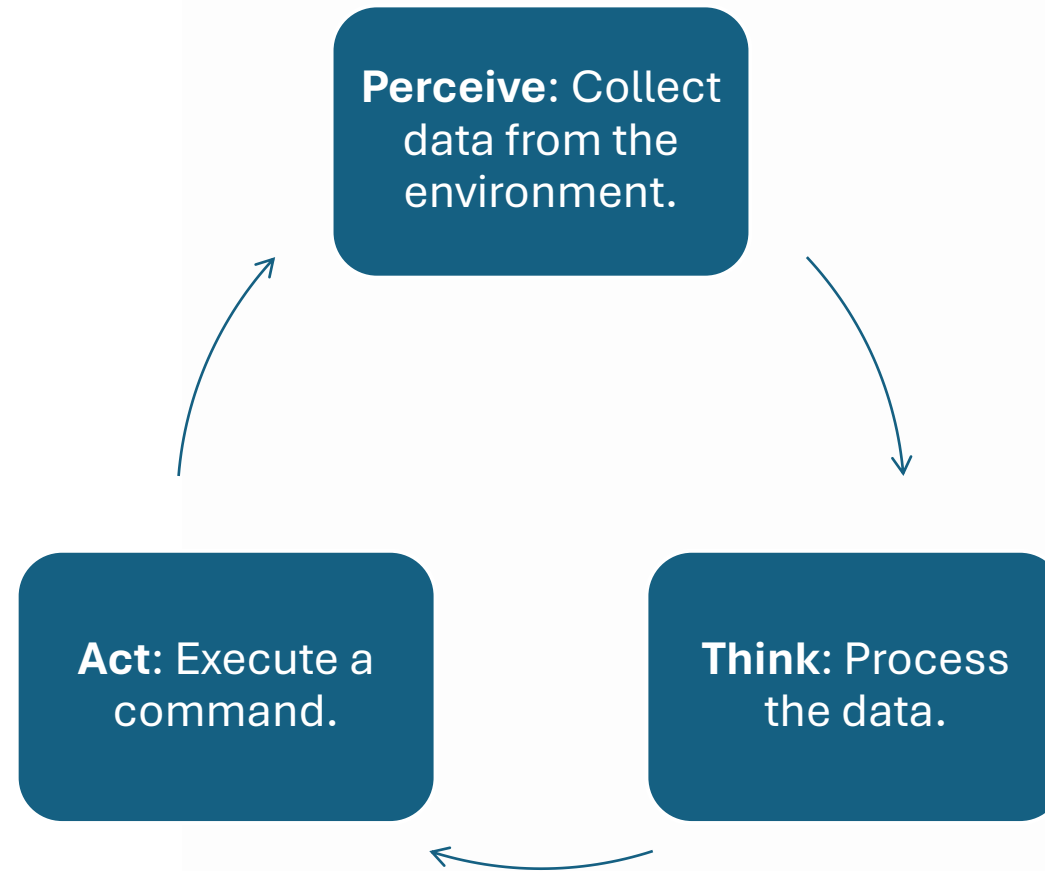
MSc (CS)[R], BSc (CS)

What is an “Agent”?

- An Agent is anything that perceives its environment and acts upon it.
- **Sensors:** Receives Data
- **Actuators:** Interacts with the environment



The Perception-Action Cycle



Bots

- A "Bot" is an agent that exists purely in software.
 - **Perceive:** Reads text/clicks/get API data.
 - **Think:** Processes logic/rules.
 - **Act:** Sends messages/updates databases/displays info.

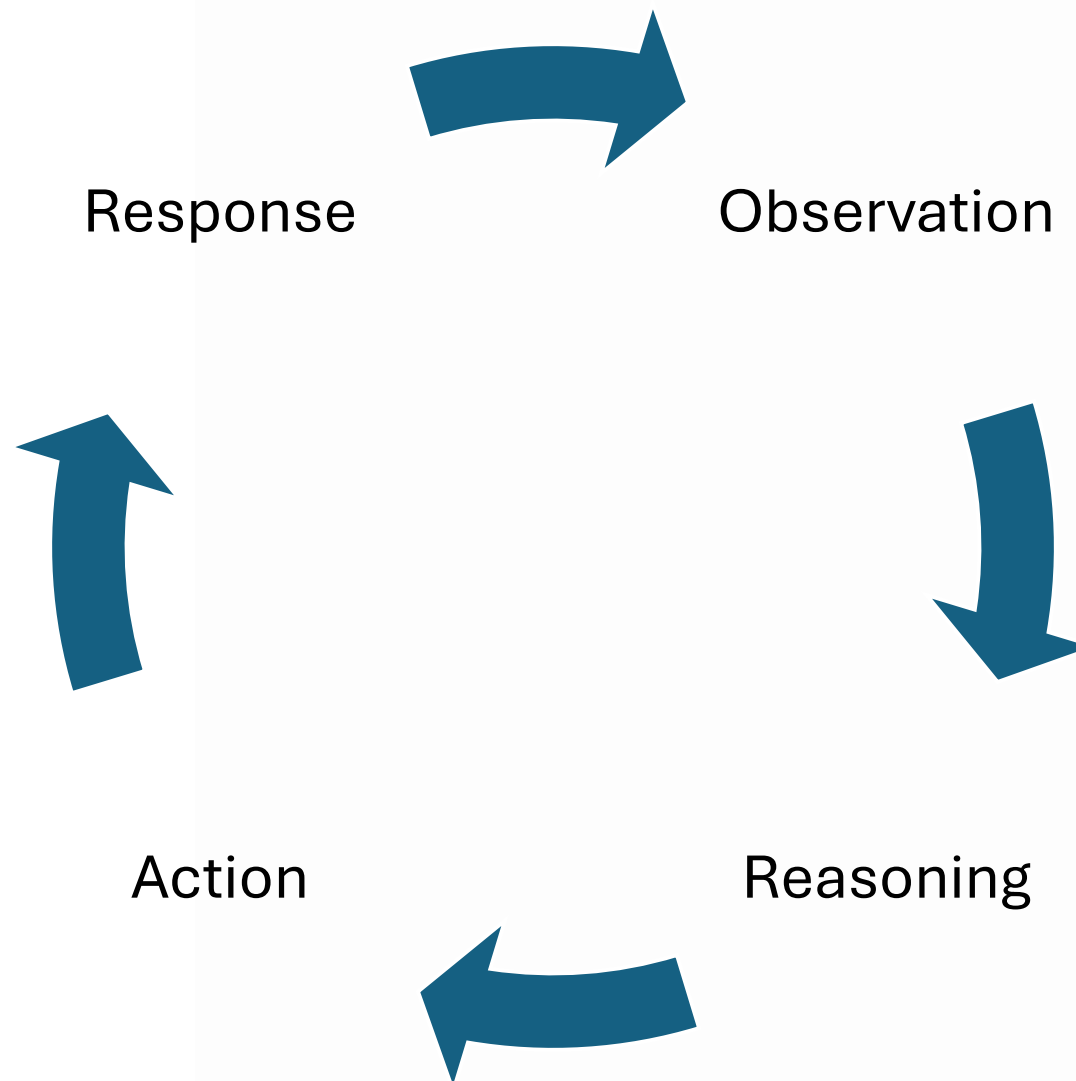


The Intent-Mapping

- Bot design is about mapping user 'Intent' to a specific 'Action.' It is the simplest form of an agent's 'Think' process.
 - **Intent:** What is the user trying to do?/What the user wants it to do?
 - **Mapping:** What is the correct response for that intent?



The Agentic Loop



Phases of the Agentic Loop

Phase	Description	Python Representation
Observation	Agent receives a user prompt.	<code>prompt = input()</code>
Reasoning	Agent checks its instructions/tools.	<code>self._model.generate(prompt)</code>
Action	Agent calls the specific tool.	<code>self._tool.execute()</code>
Persistence	Agent logs result to history.	<code>self._memory.append(...)</code>

Practical

- Go to aistudio.google.com.
- Sign in with your Google account.
- Click "Get API Key" on the left sidebar.
- Click "Create API key in new project."
- Copy this key and save it. (Never push this to GitHub).



.env file & Installing Dependencies

- .env file = “A .env file stores environment-specific configuration variables, often including sensitive data like API keys, that an application loads at runtime.”

```
pip install google-generativeai python-dotenv
```

Note: This is now depreciated (05/05/26)

Exercise

- Use the Code in <https://github.com/tobijay666/DS205.3Practicals>, to continue the activity.

